

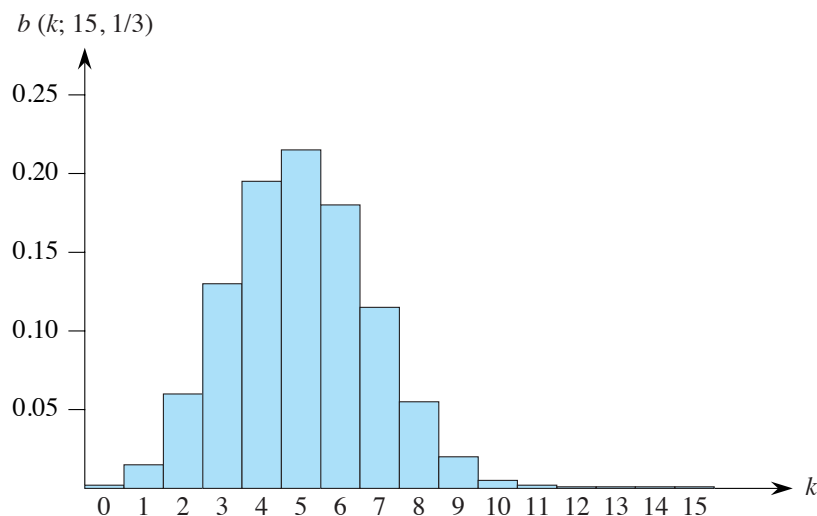


Figure C.2 The binomial distribution $b(k; 15, 1/3)$ resulting from $n = 15$ Bernoulli trials, each with probability $p = 1/3$ of success. The expectation of the distribution is $np = 5$.

$$\begin{aligned}
 E[X] &= \sum_{k=0}^n k \cdot \Pr\{X = k\} \\
 &= \sum_{k=0}^n k \cdot b(k; n, p) \\
 &= \sum_{k=1}^n k \binom{n}{k} p^k q^{n-k} \\
 &= np \sum_{k=1}^n \binom{n-1}{k-1} p^{k-1} q^{n-k} \quad (\text{by equation (C.9) on page 1183}) \\
 &= np \sum_{k=0}^{n-1} \binom{n-1}{k} p^k q^{(n-1)-k} \\
 &= np \sum_{k=0}^{n-1} b(k; n-1, p) \\
 &= np \quad (\text{by equation (C.40)}) . \tag{C.41}
 \end{aligned}$$

Linearity of expectation produces the same result with substantially less algebra. Let X_i be the random variable describing the number of successes in the i th trial. Then $E[X_i] = p \cdot 1 + q \cdot 0 = p$, and the expected number of successes for n trials is

$$\begin{aligned}
E[X] &= E\left[\sum_{i=1}^n X_i\right] \\
&= \sum_{i=1}^n E[X_i] \quad (\text{by equation (C.24) on page 1192}) \\
&= \sum_{i=1}^n p \\
&= np.
\end{aligned} \tag{C.42}$$

We can use the same approach to calculate the variance of the distribution. By equation (C.31), $\text{Var}[X_i] = E[X_i^2] - E^2[X_i]$. Since X_i takes on only the values 0 and 1, we have $X_i^2 = X_i$, which implies $E[X_i^2] = E[X_i] = p$. Hence,

$$\text{Var}[X_i] = p - p^2 = p(1 - p) = pq. \tag{C.43}$$

To compute the variance of X , we take advantage of the independence of the n trials. By equation (C.33), we have

$$\begin{aligned}
\text{Var}[X] &= \text{Var}\left[\sum_{i=1}^n X_i\right] \\
&= \sum_{i=1}^n \text{Var}[X_i] \\
&= \sum_{i=1}^n pq \\
&= npq.
\end{aligned} \tag{C.44}$$

As Figure C.2 shows, the binomial distribution $b(k; n, p)$ increases with k until it reaches the mean np , and then it decreases. To prove that the distribution always behaves in this manner, examine the ratio of successive terms:

$$\begin{aligned}
\frac{b(k; n, p)}{b(k-1; n, p)} &= \frac{\binom{n}{k} p^k q^{n-k}}{\binom{n}{k-1} p^{k-1} q^{n-k+1}} \\
&= \frac{n! (k-1)! (n-k+1)! p}{k! (n-k)! n! q} \\
&= \frac{(n-k+1)p}{kq} \\
&= 1 + \frac{(n-k+1)p - kq}{kq} \\
&= 1 + \frac{(n-k+1)p - k(1-p)}{kq}
\end{aligned} \tag{C.45}$$

$$= 1 + \frac{(n+1)p - k}{kq}.$$

This ratio is greater than 1 precisely when $(n+1)p - k$ is positive. Consequently, $b(k; n, p) > b(k-1; n, p)$ for $k < (n+1)p$ (the distribution increases), and $b(k; n, p) < b(k-1; n, p)$ for $k > (n+1)p$ (the distribution decreases). If $(n+1)p$ is an integer, then for $k = (n+1)p$, the ratio $b(k; n, p)/b(k-1; n, p)$ equals 1, so that $b(k; n, p) = b(k-1; n, p)$. In this case, the distribution has two maxima: at $k = (n+1)p$ and at $k-1 = (n+1)p-1 = np-q$. Otherwise, it attains a maximum at the unique integer k that lies in the range $np-q < k < (n+1)p$.

The following lemma provides an upper bound on the binomial distribution.

Lemma C.1

Let $n \geq 0$, let $0 < p < 1$, let $q = 1 - p$, and let $0 \leq k \leq n$. Then

$$b(k; n, p) \leq \left(\frac{np}{k}\right)^k \left(\frac{nq}{n-k}\right)^{n-k}.$$

Proof We have

$$\begin{aligned} b(k; n, p) &= \binom{n}{k} p^k q^{n-k} \\ &\leq \left(\frac{n}{k}\right)^k \left(\frac{n}{n-k}\right)^{n-k} p^k q^{n-k} \quad (\text{by inequality (C.7) on page 1182}) \\ &= \left(\frac{np}{k}\right)^k \left(\frac{nq}{n-k}\right)^{n-k}. \end{aligned}$$

■

Exercises

C.4-1

Verify axiom 2 of the probability axioms for the geometric distribution.

C.4-2

How many times on average do you need to flip six fair coins before obtaining three heads and three tails?

C.4-3

Show that the variance of the geometric distribution is q/p^2 . (*Hint*: Use Exercise A.1-6 on page 1144.)

C.4-4

Show that $b(k; n, p) = b(n-k; n, q)$, where $q = 1 - p$.

C.4-5

Show that the value of the maximum of the binomial distribution $b(k; n, p)$ is approximately $1/\sqrt{2\pi npq}$, where $q = 1 - p$.

★ C.4-6

Show that the probability of no successes in n Bernoulli trials, each with probability $p = 1/n$ of success, is approximately $1/e$. Show that the probability of exactly one success is also approximately $1/e$.

★ C.4-7

Professor Rosencrantz flips a fair coin n times, and so does Professor Guildenstern. Show that the probability that they get the same number of heads is $\binom{2n}{n}/4^n$. (*Hint:* For Professor Rosencrantz, call a head a success, and for Professor Guildenstern, call a tail a success.) Use your argument to verify the identity

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}.$$

★ C.4-8

Show that for $0 \leq k \leq n$,

$$b(k; n, 1/2) \leq 2^{n H(k/n) - n},$$

where $H(x)$ is the entropy function (C.8) on page 1182.

★ C.4-9

Consider n Bernoulli trials, where for $i = 1, 2, \dots, n$, the i th trial has probability p_i of success, and let X be the random variable denoting the total number of successes. Let $p \geq p_i$ for all $i = 1, 2, \dots, n$. Prove that for $1 \leq k \leq n$,

$$\Pr\{X < k\} \geq \sum_{i=0}^{k-1} b(i; n, p).$$

★ C.4-10

Let X be the random variable for the total number of successes in a set A of n Bernoulli trials, where the i th trial has a probability p_i of success, and let X' be the random variable for the total number of successes in a second set A' of n Bernoulli trials, where the i th trial has a probability $p'_i \geq p_i$ of success. Prove that for $0 \leq k \leq n$,

$$\Pr\{X' \geq k\} \geq \Pr\{X \geq k\}.$$

(*Hint:* Show how to obtain the Bernoulli trials in A' by an experiment involving the trials of A , and use the result of Exercise C.3-7.)

★ C.5 The tails of the binomial distribution

The probability of having at least, or at most, k successes in n Bernoulli trials, each with probability p of success, is often of more interest than the probability of having exactly k successes. In this section, we investigate the *tails* of the binomial distribution: the two regions of the distribution $b(k; n, p)$ that are far from the mean np . We'll prove several important bounds on (the sum of all terms in) a tail.

We first provide a bound on the right tail of the distribution $b(k; n, p)$. To determine bounds on the left tail, simply invert the roles of successes and failures.

Theorem C.2

Consider a sequence of n Bernoulli trials, where success occurs with probability p . Let X be the random variable denoting the total number of successes. Then for $0 \leq k \leq n$, the probability of at least k successes is

$$\begin{aligned} \Pr\{X \geq k\} &= \sum_{i=k}^n b(i; n, p) \\ &\leq \binom{n}{k} p^k. \end{aligned}$$

Proof For $S \subseteq \{1, 2, \dots, n\}$, let A_S denote the event that the i th trial is a success for every $i \in S$. Since $\Pr\{A_S\} = p^{|S|}$, where $|S| = k$, we have

$$\begin{aligned} \Pr\{X \geq k\} &= \Pr\{\text{there exists } S \subseteq \{1, 2, \dots, n\} : |S| = k \text{ and } A_S\} \\ &= \Pr\left\{\bigcup_{S \subseteq \{1, 2, \dots, n\} : |S|=k} A_S\right\} \\ &\leq \sum_{S \subseteq \{1, 2, \dots, n\} : |S|=k} \Pr\{A_S\} \quad (\text{by inequality (C.21) on page 1190}) \\ &= \binom{n}{k} p^k. \end{aligned} \quad \blacksquare$$

The following corollary restates the theorem for the left tail of the binomial distribution. In general, we'll leave it to you to adapt the proofs from one tail to the other.

Corollary C.3

Consider a sequence of n Bernoulli trials, where success occurs with probability p . If X is the random variable denoting the total number of successes, then for $0 \leq k \leq n$, the probability of at most k successes is

$$\begin{aligned}
\Pr\{X \leq k\} &= \sum_{i=0}^k b(i; n, p) \\
&\leq \binom{n}{n-k} (1-p)^{n-k} \\
&= \binom{n}{k} (1-p)^{n-k}.
\end{aligned}$$

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Our next bound concerns the left tail of the binomial distribution. Its corollary shows that, far from the mean, the left tail diminishes exponentially.

Theorem C.4

Consider a sequence of n Bernoulli trials, where success occurs with probability p and failure with probability $q = 1 - p$. Let X be the random variable denoting the total number of successes. Then for $0 < k < np$, the probability of fewer than k successes is

$$\begin{aligned}
\Pr\{X < k\} &= \sum_{i=0}^{k-1} b(i; n, p) \\
&< \frac{kq}{np - k} b(k; n, p).
\end{aligned}$$

Proof We bound the series $\sum_{i=0}^{k-1} b(i; n, p)$ by a geometric series using the technique from Section A.2, page 1147. For $i = 1, 2, \dots, k$, equation (C.45) gives

$$\begin{aligned}
\frac{b(i-1; n, p)}{b(i; n, p)} &= \frac{iq}{(n-i+1)p} \\
&< \frac{iq}{(n-i)p} \\
&\leq \frac{kq}{(n-k)p}.
\end{aligned}$$

If we let

$$\begin{aligned}
x &= \frac{kq}{(n-k)p} \\
&< \frac{kq}{(n-np)p} \\
&= \frac{kq}{nqp}
\end{aligned}$$

$$\begin{aligned}
&= \frac{k}{np} \\
&< 1,
\end{aligned}$$

it follows that

$$b(i-1; n, p) < x b(i; n, p)$$

for $0 < i \leq k$. Iteratively applying this inequality $k-i$ times gives

$$b(i; n, p) < x^{k-i} b(k; n, p)$$

for $0 \leq i < k$, and hence

$$\begin{aligned}
\sum_{i=0}^{k-1} b(i; n, p) &< \sum_{i=0}^{k-1} x^{k-i} b(k; n, p) \\
&< b(k; n, p) \sum_{i=1}^{\infty} x^i \\
&= \frac{x}{1-x} b(k; n, p) \\
&= \frac{kq / ((n-k)p)}{((n-k)p - kq) / ((n-k)p)} b(k; n, p) \\
&= \frac{kq}{np - kp - kq} b(k; n, p) \\
&= \frac{kq}{np - k} b(k; n, p). \quad \blacksquare
\end{aligned}$$

Corollary C.5

Consider a sequence of n Bernoulli trials, where success occurs with probability p and failure with probability $q = 1 - p$. Then for $0 < k \leq np/2$, the probability of fewer than k successes is less than half the probability of fewer than $k + 1$ successes.

Proof Because $k \leq np/2$, we have

$$\begin{aligned}
\frac{kq}{np - k} &\leq \frac{(np/2)q}{np - (np/2)} \\
&= \frac{(np/2)q}{np/2} \\
&\leq 1,
\end{aligned} \tag{C.46}$$

since $q \leq 1$. Letting X be the random variable denoting the number of successes, Theorem C.4 and inequality (C.46) imply that the probability of fewer than k successes is

$$\Pr\{X < k\} = \sum_{i=0}^{k-1} b(i; n, p) < b(k; n, p) .$$

Thus we have

$$\begin{aligned} \frac{\Pr\{X < k\}}{\Pr\{X < k + 1\}} &= \frac{\sum_{i=0}^{k-1} b(i; n, p)}{\sum_{i=0}^k b(i; n, p)} \\ &= \frac{\sum_{i=0}^{k-1} b(i; n, p)}{\sum_{i=0}^{k-1} b(i; n, p) + b(k; n, p)} \\ &< 1/2 \end{aligned}$$

since $\sum_{i=0}^{k-1} b(i; n, p) < b(k; n, p)$. ■

Bounds on the right tail follow similarly. Exercise C.5-2 asks you to prove them.

Corollary C.6

Consider a sequence of n Bernoulli trials, where success occurs with probability p . Let X be the random variable denoting the total number of successes. Then for $np < k < n$, the probability of more than k successes is

$$\begin{aligned} \Pr\{X > k\} &= \sum_{i=k+1}^n b(i; n, p) \\ &< \frac{(n-k)p}{k-np} b(k; n, p) . \end{aligned} \quad \blacksquare$$

Corollary C.7

Consider a sequence of n Bernoulli trials, where success occurs with probability p and failure with probability $q = 1 - p$. Then for $(np + n)/2 < k < n$, the probability of more than k successes is less than half the probability of more than $k - 1$ successes. ■

The next theorem considers n Bernoulli trials, each with a probability p_i of success, for $i = 1, 2, \dots, n$. As the subsequent corollary shows, we can use the theorem to provide a bound on the right tail of the binomial distribution by setting $p_i = p$ for each trial.

Theorem C.8

Consider a sequence of n Bernoulli trials, where in the i th trial, for $i = 1, 2, \dots, n$, success occurs with probability p_i and failure occurs with probability $q_i = 1 - p_i$. Let X be the random variable describing the total number of successes, and let $\mu = E[X]$. Then for $r > \mu$,

$$\Pr\{X - \mu \geq r\} \leq \left(\frac{\mu e}{r}\right)^r.$$

Proof Since for any $\alpha > 0$, the function $e^{\alpha x}$ strictly increases in x ,

$$\Pr\{X - \mu \geq r\} = \Pr\{e^{\alpha(X-\mu)} \geq e^{\alpha r}\}, \quad (\text{C.47})$$

where we will determine α later. Using Markov's inequality (C.34), we obtain

$$\Pr\{e^{\alpha(X-\mu)} \geq e^{\alpha r}\} \leq \mathbb{E}[e^{\alpha(X-\mu)}] e^{-\alpha r}. \quad (\text{C.48})$$

The bulk of the proof consists of bounding $\mathbb{E}[e^{\alpha(X-\mu)}]$ and substituting a suitable value for α in inequality (C.48). First, we evaluate $\mathbb{E}[e^{\alpha(X-\mu)}]$. Using the technique of indicator random variables (see Section 5.2), let $X_i = \mathbb{I}\{\text{the } i\text{th Bernoulli trial is a success}\}$ for $i = 1, 2, \dots, n$. That is, X_i is the random variable that is 1 if the i th Bernoulli trial is a success and 0 if it is a failure. Thus, we have

$$X = \sum_{i=1}^n X_i,$$

and by linearity of expectation,

$$\mu = \mathbb{E}[X] = \mathbb{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbb{E}[X_i] = \sum_{i=1}^n p_i,$$

which implies

$$X - \mu = \sum_{i=1}^n (X_i - p_i).$$

To evaluate $\mathbb{E}[e^{\alpha(X-\mu)}]$, we substitute for $X - \mu$, obtaining

$$\begin{aligned} \mathbb{E}[e^{\alpha(X-\mu)}] &= \mathbb{E}[e^{\alpha \sum_{i=1}^n (X_i - p_i)}] \\ &= \mathbb{E}\left[\prod_{i=1}^n e^{\alpha(X_i - p_i)}\right] \\ &= \prod_{i=1}^n \mathbb{E}[e^{\alpha(X_i - p_i)}], \end{aligned}$$

which follows from equation (C.27), since the mutual independence of the random variables X_i implies the mutual independence of the random variables $e^{\alpha(X_i - p_i)}$ (see Exercise C.3-5). By the definition of expectation,

$$\begin{aligned}
\mathbb{E}[e^{\alpha(X_i - p_i)}] &= e^{\alpha(1-p_i)} p_i + e^{\alpha(0-p_i)} q_i \\
&= p_i e^{\alpha q_i} + q_i e^{-\alpha p_i} \\
&\leq p_i e^{\alpha} + 1 \\
&\leq \exp(p_i e^{\alpha}),
\end{aligned} \tag{C.49}$$

where $\exp(x)$ denotes the exponential function: $\exp(x) = e^x$. (Inequality (C.49) follows from the inequalities $\alpha > 0$, $q_i \leq 1$, $e^{\alpha q_i} \leq e^{\alpha}$, and $e^{-\alpha p_i} \leq 1$. The last line follows from inequality (3.14) on page 66.) Consequently,

$$\begin{aligned}
\mathbb{E}[e^{\alpha(X - \mu)}] &= \prod_{i=1}^n \mathbb{E}[e^{\alpha(X_i - p_i)}] \\
&\leq \prod_{i=1}^n \exp(p_i e^{\alpha}) \\
&= \exp\left(\sum_{i=1}^n p_i e^{\alpha}\right) \\
&= \exp(\mu e^{\alpha}),
\end{aligned} \tag{C.50}$$

since $\mu = \sum_{i=1}^n p_i$. Therefore, from equation (C.47) and inequalities (C.48) and (C.50), it follows that

$$\Pr\{X - \mu \geq r\} \leq \exp(\mu e^{\alpha} - \alpha r). \tag{C.51}$$

Choosing $\alpha = \ln(r/\mu)$ (see Exercise C.5-7), we obtain

$$\begin{aligned}
\Pr\{X - \mu \geq r\} &\leq \exp(\mu e^{\ln(r/\mu)} - r \ln(r/\mu)) \\
&= \exp(r - r \ln(r/\mu)) \\
&= \frac{e^r}{(r/\mu)^r} \\
&= \left(\frac{\mu e}{r}\right)^r.
\end{aligned}$$

■

When applied to Bernoulli trials in which each trial has the same probability of success, Theorem C.8 yields the following corollary bounding the right tail of a binomial distribution.

Corollary C.9

Consider a sequence of n Bernoulli trials, where in each trial success occurs with probability p and failure occurs with probability $q = 1 - p$. Then for $r > np$,